

Baseera

VOICE GUIDED - OBJECT FINDER

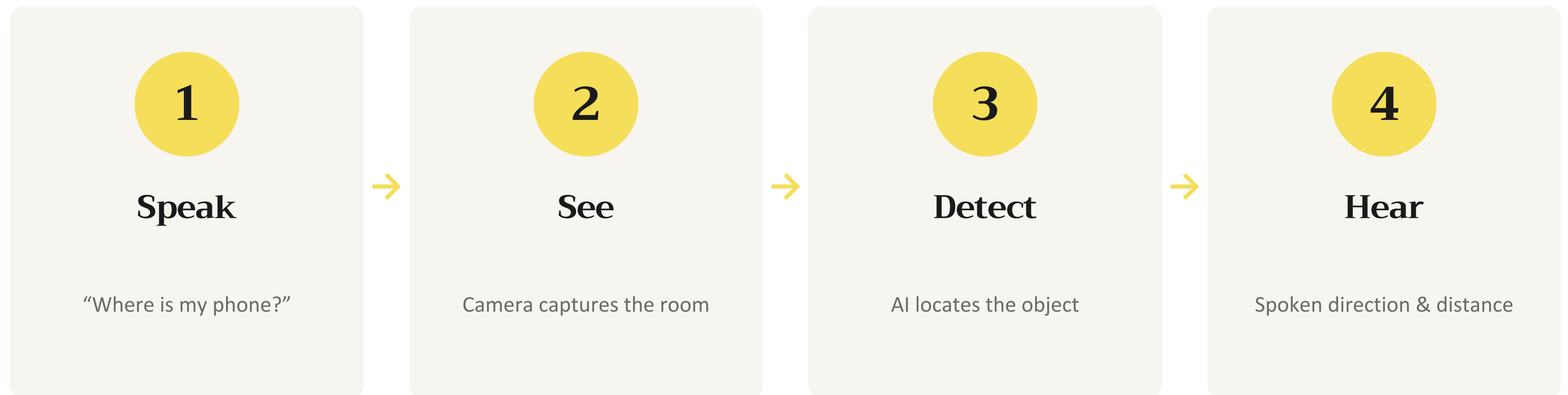
The Problem

Millions of visually impaired people rely on touch and memory to find everyday objects — a phone, a bottle, their keys. Simple tasks become slow and frustrating without sight.

- No affordable, real-time way to locate objects by voice
- Existing tools are either too costly or English-only
- Assistance often depends on another person being present

The Solution

Ask out loud. Baseera looks, finds, and tells you exactly where it is.



How It Works

1	Whisper (STT)	Transcribes the spoken question in Arabic or English
2	Qwen2.5 LLM	Extracts the target object name from the question
3	YOLOv8n + YOLOv8s	Two detectors scan the photo; highest-confidence match wins
4	Geometry	Bounding box position & size give direction and distance
5	Qwen2.5 LLM	Composes one natural spoken sentence, in the user's language
6	gTTS	Converts the sentence to speech and plays it back

Key Features

1 Bilingual by design

Understands and replies naturally in Arabic and English, auto-detected.

2 Dual-model detection

Runs YOLOv8n and YOLOv8s together and keeps the most confident match.

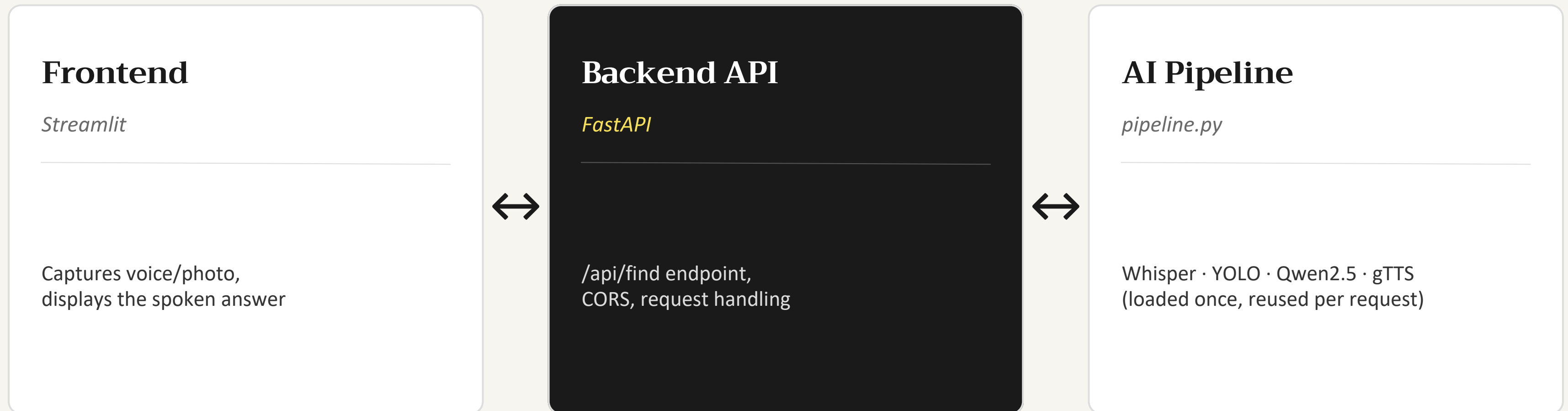
3 Distance & direction

Pinhole-camera math turns a bounding box into meters and a clock-style direction.

4 Safe language fallback

A template sentence steps in if the LLM's reply isn't trustworthy.

System Architecture



HTTP requests carry image + audio/text; JSON response carries text and base64 audio.

Team & Responsibilities

P1

Speech & Language

Backend API

P2

Vision & Detection

Frontend UI

P3

Response & Voice

Containerization

P4

Orchestration & Testing

Cloud & Mobile

Deployment & What's Next

Today

- FastAPI backend + Streamlit frontend, run locally or in Docker
- Colab + ngrok tunnel for quick mobile demos
- Both YOLO checkpoints and the LLM load once at startup

Next

- Persistent hosting (no more resetting ngrok links)
- Calibrate distance formula per real device camera
- Expand the object vocabulary beyond common household items
- On-device / offline mode for low-connectivity use



Baseera

Thank You

Questions & Discussion